

Muhammad Khalid

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WORK EXPERIENCE

Founding Software Engineer

Sept. 2025 – Sept. 2026

Coco Health

San Francisco, CA

- Owned the **Python/LiveKit** voice runtime and core backend services across a **TypeScript/NestJS** codebase, spanning agent orchestration, model infrastructure, tool execution and production reliability
- Increased average daily conversations by **100%**, by designing and shipping an autonomous **agent scheduling and task-execution system** for proactive user follow-ups
- Built durable **agent context and memory systems** spanning persistent conversation history, summarization, per-turn context assembly, and failure recovery
- Built a multi-provider voice pipeline with **OpenTelemetry** observability, provider fallback, turn-latency metrics; later migrated the core Voice AI LLM/TTS architecture to reduce average TTFT by **60%** from **~2.5s** to **~1s** and end-to-end response latency by **~1s**
- Increased notification open rates from **~5%** to **~10%** by redesigning personalized health news research from a deterministic pipeline into an agentic research workflow
- Built **file** and **document-processing infrastructure** with signed **GCS** uploads, asynchronous PDF parsing/OCR, processing state machines, and extracted content surfaced to agents at runtime

Lead Developer, PioneerMart

January 2025 – May 2026

Grinnell College

Grinnell, IA

- Led engineering team on a **React Native + Django** marketplace app, deploying **20+ full-stack features** with **PostgreSQL**, **Zustand** (state management), **Redis**, and **WebSockets**, used by **300+ students** in **week 1**
- Engineered production deployment with **Nginx** reverse proxy and **Daphne** on a **Linux**-based cloud provider, enabling stable real-time functionality
- Integrated **OTP** auth, image moderation (Sightengine API), and secure file storage via **AWS S3**

Research Assistant

May 2025 – August 2025

Grinnell College

Grinnell, IA

- Engineered an OCR-LLM benchmarking pipeline across multiple mLLMs on **400+** historical text pages
- Standardized OCR post-correction workflow across a team of 5

PROJECTS

Open-Source Contributions | *LlamaIndex, FastMCP*

July 2025

- Merged [PR #19439](#) in the **LlamaIndex** codebase (**43k+ GitHub stars**) fixing a triaged bug in the **llama-dev CLI** test framework, improving developer feedback on Python packages
- Merged [PR #1257](#) in the **FastMCP** codebase (**27.7k+ GitHub stars**) fixing JSON Schema generation

Heartbeat Time-Series Classification | *PyTorch, InceptionTime, scikit-learn*

December 2025

- Trained and evaluated an **InceptionTime CNN** for normal-vs.-abnormal heartbeat time-series classification, using per-signal normalization, class-weighted cross-entropy, and L2 regularization; achieved **86.6%** test accuracy and **0.868** weighted F1.

EDUCATION

Grinnell College

Grinnell, IA

B.A. in Computer Science

2022 - 2026

- Cumulative GPA: 3.7/4.0, In-major GPA: 3.75/4.0, Deans list all semesters
- Selected Coursework: Computational Methods, Software Design & Development, Operating Systems & Parallel Algorithms, Data Structures & Algorithms, Object-Oriented Programming, Linear Algebra

TECHNICAL SKILLS

Languages: TypeScript, Python, JavaScript, Java

Backend & Distributed Systems: Node.js, NestJS, GraphQL, BullMQ, Django, Livekit

Data & Infrastructure: MongoDB, PostgreSQL, Redis, GCP, AWS, OpenTelemetry

AI/ML: MCP, PyTorch, scikit-learn

Certifications: AWS Certified Solutions Architect – Associate